

Arpit Mandal

Software Developer (4 Years+) | Full Stack Engineer | Victoria, BC (Open to Relocation)
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EXPERIENCE

Reliable Controls Corp. | Software Developer L1

Victoria, British Columbia, CA | (May 2025 – Currently working)

- As part of the R&D portal team, collaborated with platform, engineering, and core teams to migrate a WebView app from DDD (Onion) to N-Tier Clean Architecture on .NET 10, using EF Core, Async/Await, and Dependency Injection, simplified LINQ queries, added indexing strategies, and validated through performance benchmarking, achieving ~35% gains over the legacy system.
- Built and maintained microservice-based integrations (WCF + HTTP) across the RC ecosystem, implementing SMTP, SSO, and SAML authentication and services such as Auth, Device COMM, TSDE etc. and worked with building-automation protocols (BACnet, IPv4/IPv6, BSN). Integrated NUnit and JMeter for unit, integration automated testing, with structured DB logging for reliability. Performed code reviewed and AI assisted development
- Designed and maintained CI/CD pipelines via Azure DevOps, managed Git submodules, created automated installers pipelines, and supported production debugging and contributed to PI-release Functional Stress Testing, benchmark calculations, and Innovation Day projects, driving cost savings across development, testing, and deployment. Participated and provided inputs during scrum ceremonies and SAT discussions.
- Participated in Agile (Scrum) ceremonies and SR&ED meetings; documented to Confluence, supported cross-team knowledge sharing across portal, platform, and core teams, took end-to-end ownership of the WebView migration, from architecture discussions and design decisions through to scalable, production-ready delivery.

Cognizant Technology Solutions | GenAI and Azure (Externship)

Toronto, Ontario, CA | (Jan 2025 - March 2025)

- Completed structured training in Generative AI, Prompt Engineering, LLM fine-tuning, and Azure AI services (Azure Functions, Cognitive Services, Azure OpenAI), applied concepts hands-on in cloud-native, AI assisted and AI driven development.
- Designed and developed an AI-integrated full-stack application using .NET, Razor Pages, and Gemini API, implementing LLM-powered features end-to-end including API integration, structured data persistence, and scalable cloud-ready application design, also built a Capstone Agentic Chatbot leveraging Azure OpenAI and Azure Functions, demonstrating production-oriented GenAI integration patterns.

Cognizant Technology Solutions | Associate Full Stack Developer

Chennai, India | (Oct 2020 - July 2023)

- Spearheaded development and core developer for Toyota Motors manufacturing plant product, working closely with dev, QA, DevOps, and client stakeholders to modernize a legacy on-prem system into a production-grade .NET Core + Angular application which is architected and deployed on AWS using Terraform IaC (EC2, S3, IAM, Lambda, ECS), delivering ~28% improvement in KPIs, throughput, and response time over the legacy baseline. Implemented indexing and rollover logging frameworks for automated cleanups, wrote unit testable code.
- Designed and implemented RESTful APIs and microservices architecture supporting event-driven workflows including job scheduling using hangfire.io and real-time data processing which helped us to enable seamless integration across distributed enterprise systems.
- Built full-stack, production-ready features using C# / .NET 8, EF Core, LINQ, Async/Await, Dependency Injection, and Angular, optimized SQL Server queries, introduced reusable component patterns, and maintained high standards for performance, UI/UX responsiveness, and long-term maintainability across sprints.
- Strengthened the DevOps delivery pipeline by building and maintaining CI/CD pipelines with Jenkins and Docker; automated deployments, integrated test automation, and supported production releases, actively participated in Agile (Scrum) ceremonies, performed code reviews, led design discussions, and handled triage across multiple release cycles.

EDUCATION

George Brown College, Toronto, ON – Post Graduation in Cloud Computing

Relevant courses: Cloud Fundamentals, Azure, AWS, DevOps, Cloud Security, Networking, Architecture, Hyper-V, Virtualization

DIT University, Dehradun, IN - Bachelor of Engineering in CS

Relevant courses: C, Java, C#, DSA, OOPS, System Design, Cryptography and Encryption, Cloud, Python, Microprocessor, RDBMS

SKILLS

- Languages & Frameworks:** C#, .NET 10 (ASP.NET Core, MVC, Razor Pages, EF Core, LINQ, Async/Await, WCF, Dependency Injection, Hangfire), Python (Django), JavaScript/TypeScript (Angular), RESTful APIs, Microservices, OOP, XML
- Cloud, DevOps & Infrastructure:** AWS (EC2, S3, IAM, Lambda, ECS), Azure (Functions, Cognitive Services, Azure DevOps), Docker, Jenkins, Terraform, CloudFormation, CI/CD Pipelines, Git, Git Submodules, Hyper-V, Virtualization
- Databases & Testing:** SQL Server, MySQL, SQLite, Query Optimization, Indexing, NUnit, Moq, JMeter, Postman, Swagger, Log4Net
- AI/ML, Practices & Tools:** Gemini API, Azure OpenAI, Azure Cognitive Services, Ollama (LLaMA), Prompt Engineering, LLM Integration, System Design, OOD, DSA, Agile (Scrum/Kanban), SAML, SSO, BACnet, Confluence, Jira
- Domains Worked:** Manufacturing and Automobile, Pharmacy, Building automation (HVAC and BAS), Pharmaceutical

SELECTED PROJECTS

- ATS Resume Analyser & Cover Letter Generator:** Built a full-stack .NET + Razor Pages application powered by Gemini API (LLM) with three core modules: Resume Manager (upload, role-fit analysis, AI improvement suggestions), ATS Analyser (JD via URL or paste, resume-JD alignment scoring), and Cover Letter Generator (configurable template editor, live preview, download). Persisted all data to SQL Server; packaged as a desktop-installable app running a local web server that communicates with the Gemini AI API, delivering a fully self-contained, configurable AI-driven tool.
- Electronic Shift Log (Toyota Motors):** Re-engineered a legacy manufacturing system into a .NET Core + Angular app with Clean Architecture, deployed on AWS (EC2, S3, IAM) via Terraform, hence achieved ~28% KPI performance improvement through cloud-native integration and reusable API design and angular UI components.
- WebView Migration (Reliable Controls):** Led migration initiative of a Windows/Web-based WebView app to N-Tier Clean Architecture on .NET 10 with EF Core, and modern technologies, benchmarked ~35% performance gains over the legacy system through query optimization and indexing.
- Vehicle Damage Cost Predictor:** Built a computer vision prediction model prototype at a Toyota Motors NA hackathon, winning under competitive conditions; applied ML techniques for automated vehicle damage detection and repair cost estimation.

ACHIEVEMENTS / INTERESTS

- Toyota Motors NA hackathon:** built a vehicle damage cost prediction prototype using a computer vision ML model under competitive conditions.
- Built an Innovation Day prototype at Reliable Controls using local LLM - Ollama + LLaMA 3.2 (3B) to summarize building alarms and surface high-priority alerts on the user dashboard at login.
- Solving Leet-code problems and contributing to Open-source GitHub projects.